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A paradigm shift: Low-cost sensors for effective air quality monitoring and management in developing countries



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ABSTRACT

This study gives an overview of the air quality monitoring challenges faced by South Asian countries, with a specific focus on Pakistan, and explores the potential application of low-cost sensors (LCS) to address these issues. Currently, 89% of the 4.2 million premature global deaths attributable to ambient air pollution occur in low- and middle-income countries, underscoring the urgent need for improved monitoring and abatement measures. In Pakistan, these challenges result in significant public health and economic consequences due to institutional and financial constraints, limited data availability, and transboundary pollution. The situation is exacerbated by the absence of an effective air quality regulatory network. This study proposes a shift by establishing a hybrid monitoring network that integrates conventional regulatory instruments with LCS. The current PM_{2.5} pollution scenario in major Pakistani cities is analyzed using data from 61 LCSs installed nationwide, with one TSI BlueSky sensor validated against a regulatory BAM (Beta Attenuation Monitor) in Chakwal. Results show that the unadjusted BlueSky values demonstrate a strong correlation (Pearson correlation 0.85) with the reference BAM instrument, with acceptable error margins (NRMSE and NMAE of 8% and 6%, respectively), indicating the sensor's potential for reliable monitoring. Though adjusted values aligned better, the analysis focuses on unadjusted data for broader applicability. These findings suggest that combining LCSs with advanced data analytics can serve as a technically and economically viable solution for accurate air quality monitoring and effective management in Pakistan and other developing countries facing similar high PM_{2.5} concentrations. This paradigm shift enhances monitoring capabilities and supports informed policy-making and public health initiatives.

1. Introduction

According to the statistics published by World Health Organization (WHO), nearly all the people inhabiting Earth (99 %) breathe polluted air that surpasses WHO recommended limits, while highest exposure rates have been recorded in low- and middle-income countries (WHO, 2023). Air pollution is declared as the fourth most significant global risk contributing to mortality (World Bank, 2016) and is responsible for causing over 7 million premature deaths, every year, on a global scale (UNEP, 2021). Moreover, estimates show that 4.2 million premature casualties worldwide were ascribed to ambient (outdoor) air pollution in both urban and rural areas during 2019 (WHO, 2022). These casualties are primarily attributed to fine particulate matter exposure, which is the

major cause of cancers, cardiovascular and respiratory diseases. Particulate matter is a typical proxy marker of air pollution and may constitute; SO₄²⁻ (sulfates), NO₃⁻ (nitrates), NH₃ (ammonia), NaCl (sodium chloride, predominantly from sea salt), BC (black carbon), mineral dust (WHO, 2022) and organic aerosol, which includes both primary and secondary organic fractions (Peng et al., 2025). However, people in low- and middle-income countries bear a disproportionate amount of the burden of outdoor air pollution, as they account for 89 % of the 4.2 million premature deaths that occur globally. Notably, the highest burden is reported in the WHO Regions of South-East Asia and Western Pacific.

Air quality monitoring (AQM) is of paramount importance globally, and particularly in this region with severe implications for both human

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health and the environment. Traditionally, AQM relied on sophisticated and costly stationary monitors at fixed sites (Kularatna and Sudantha, 2008). However, the technology has evolved significantly (Snyder et al., 2013), with the advent of low-cost and user-friendly air pollution sensors that offer real-time monitoring at a reasonable cost, thus enabling greater spatial coverage compared to traditional devices. This paradigm shift is evident worldwide (Karagulian et al., 2019).

The term “low-cost” denotes the acquisition price of the sensors in comparison to the equipment and operational cost of the reference-grade monitors. The terminology is relative, with respect to the user and the intended application. In various publications (e.g., (Hall et al., 2014, Rai et al., 2017)), the low-cost air quality sensor modules or monitors range in price from several hundred to a few thousand dollars. In this study, the cost of a BlueSky low-cost sensor from TSI was \$800 USD and therefore, a sensor costing less than \$1000 USD was considered a low-cost sensor (Kang et al., 2022). Besides the BlueSky monitors come with a one-year operational lifetime that can be extended through careful maintenance and biannual calibration, as demonstrated in this study and elaborated in a companion manuscript (Saeed et al., under revision).

Recent work has highlighted the ability of calibrated low-cost PM_{2.5} sensors to accurately measure concentrations under a wide range of atmospheric conditions (Zheng et al., 2018, Johnson et al., 2018). A review of 50 low-cost sensors demonstrated high accuracy, with an R^2 value of 0.99 after calibration. For instance, *Plantower sensors*, within a PM_{2.5} concentration range of 16–75 $\mu\text{g}/\text{m}^3$, exhibited an R^2 of 0.96 and precision within 16 %, while *Novasense sensors*, within a PM_{2.5} concentration range of 0–120 $\mu\text{g}/\text{m}^3$, had an R^2 of 0.90 and precision within 12 % compared to reference instruments (Alfano et al., 2020). This data advocates that low-cost sensors (LCSs) are suitable for environmental science research. Studies employing LCSs in various environments—semi-urban, urban, and rural—have underscored the necessity of calibrating raw data from LCS with reference instruments, such as FEM (Federal Equivalent Method) or FRM (Federal Reference Method) instruments, before analysis (Raysoni et al., 2023). In this study, the *BAM (Beta Attenuation Monitor)* was used for calibration and validation, a widely used reference instrument in numerous studies (Feenstra, 2020, Stavroulas et al., 2020, Magi et al., 2020, Malings et al., 2020, Feenstra et al., 2019, Gupta et al., 2018, Levy Zamora et al., 2018, Zheng et al., 2018, Carvlin et al., 2017, Kelly et al., 2017, Jiao et al., 2016).

Additionally, studies using *PurpleAir sensors* consistently show a strong correlation between raw data and reference instruments (Wells, 2020), with an R^2 value exceeding 0.83. In a 320-day study, *Plantower sensors* demonstrated a strong correlation ($R^2 > 0.86$) with reference instruments, particularly during winter months (Sayahi et al., 2019). The performance of LCS is influenced by specific meteorological parameters, and algorithms are applied considering these conditions to minimize the interference (Huang et al., 2022). Fortunately, the quality of sensor data has improved significantly as LCS and calibration methodologies have rapidly developed (Lung et al., 2022).

Das and Ramachandran (2020) evaluated the low-cost PM sensor i.e., BlueSky Air Monitor Model 8143 *TSI Inc.* over a wide range of concentrations. A Dust Trak Model 8534 *TSI Inc.*, a popular light scattering device, was used to compare the particulate matter concentrations of PM₁₀, PM₄, PM_{2.5}, and PM₁ with reference data. The findings of the study showed that the BlueSky sensor and the reference instrument compare well ($R^2 = 0.92$; Intercept ~ 0 ; Slope = 0.75). Moreover, a recent study conducted at Colorado State University proposes that the LCSs are only efficient at characterizing the ultrafine ($d_p < 0.1 \mu\text{m}$) and fine ($d_p < 2.5 \mu\text{m}$) particulate matter and are unable to detect or classify practically all the coarse ($d_p < 2.5\text{--}10 \mu\text{m}$) particle fraction in the outdoor settings, in which these sensors are typically deployed (Rueda et al., 2023).

Given that air quality is hyper-local and can vary significantly over short distances, employing expensive air quality sensors for localized measurements is challenging due to the large number required (Veiga

et al., 2021). Low-cost sensors offer a potential solution by providing denser spatial coverage of real-time measurements, particularly in high-pollution regions such as Pakistan. The emergence of affordable and user-friendly PM_{2.5} sensors presents an opportunity to advance PM research in resource-constrained countries across the globe. Better characterization of LCS networks in countries like Pakistan is critical for effective citizen science engagement in pollution assessment, reliable air pollution monitoring, efficient traffic management, and credible health evaluations (Gerboles et al., 2017). Furthermore, it is significant for mitigating PM_{2.5} health hazards by helping policymakers prioritize source controls, identify pollution hotspots, and develop behavioral change strategies (Lung et al., 2022).

Globally, there are numerous applications of LCSs. The use of low-cost sensors enables the real-time and time-averaged measurement of pollutant concentrations, the creation of live air quality maps, personal exposure assessments, and the facilitation of citizen and community science initiatives, with online data accessibility. For instance, a recent study in Ji'nan, China, equipped 100 taxis with PM_{2.5} and PM₁₀ sensors, allowing them to serve as mobile air quality monitoring platforms. These taxis collectively cover 23,000 km daily, monitoring 95 % of the city's roads and generating 1.2 million PM data points per day. This system is expected to assist city authorities in evaluating the impact of traffic, dust, and near-road emissions on air quality, leading to more effective pollution control strategies. The Luftdaten project (<http://luftdaten.info>) also operates an active network of over 2,000 PM monitors (Morawska et al., 2018). Citi-Sense is another such project that maintained a network of more than 330 sensor platforms for air quality monitoring, providing hourly air quality updates across eight cities for over a month (Thompson, 2016). CAIRSENSE (Jiao et al., 2016) and Village Green (Jiao et al., 2015) also examples of such projects.

1.1. South Asian air pollution challenges

Air pollution has become a significant environmental concern in many South Asian countries (Anwar et al., 2021), impacting people's health and seriously disrupting socio-economic activities. Approximately 32 % of all deaths worldwide are linked to exposure to extremely high levels of poor ambient air quality in this region (Lelieveld et al., 2020). Four of the top five most polluted countries in the world are in South Asia: Bangladesh, Nepal, India, and Pakistan. According to AQLI (Air Quality Life Index) data, these countries have average life expectancy reduced by 5.6 years and account for 58 % of all life years lost because of particulate pollution levels far exceeding the WHO limits, owing to the region's large population and pollutant concentrations (Greenstone et al., 2024). According to the IQAir (2022) report, India and Pakistan encountered the most adverse air quality conditions during 2022, as Lahore topped the list as the most polluted city of the world.

Bangladesh, India, Nepal and Pakistan are located on the Indo-Gangetic Plains (IGP), where circulation, emissions, poor management and dense population all work together to constitute major air pollution challenges (Khokhar et al., 2021). Smog, now referred to as the region's “fifth season,” (Majeed et al., 2024) disrupts daily life, prompting emergency measures such as school closures and mobility restrictions. As the situation worsens, more drastic and multifaceted measures are anticipated (Khokhar et al., 2023). Addressing this crisis requires regional collaboration, yet the structural flaws along with the limited understanding of the air pollutant's composition, origins, and possible transboundary effects due to the lack of impartial AQM throughout South Asia hampers effective mitigation efforts (Khokhar et al., 2023).

1.2. Air quality management challenges in Pakistan

Air pollution in Pakistan remains a serious concern (Khwaja et al., 2013) with major cities such as Lahore, Karachi, Faisalabad, Quetta, Peshawar, and Islamabad/Rawalpindi, consistently exceeding both National Environmental Quality Standards (NEQS) i.e., 35 $\mu\text{g}/\text{m}^3$ for the

24-hour period and $15 \mu\text{g}/\text{m}^3$ annually (Pak-EPA, 2012), and WHO air quality guidelines (Niaz et al., 2016, Javed et al., 2014, Khwaja et al., 2013). High levels of $\text{PM}_{2.5}$ contribute to an estimated 105,000 premature deaths annually (Giannadaki et al., 2016). WHO estimates that annual burden of disease from outdoor air pollution results in 22,000 premature adult fatalities and 163,432 DALYs (Disability- Adjusted Life Years) lost in Pakistan (Anjum et al., 2021).

The primary contributors of air pollution are biomass burning, vehicular emissions, industry, thermal power generation and trans-boundary pollution (Gupta et al., 2017). Pollution levels tend to be higher during winters because of temperature inversion, elevated biomass burning for heating (Rasheed et al., 2015, Shahid et al., 2015, Gupta et al., 2013, Stone et al., 2010) and land clearing activities for crop rotation. Satellite data also revealed persistent poor air quality in Lahore and other cities (Khokhar et al., 2016). Ammonia concentrations are notably high in Punjab, primarily due to agricultural emissions (Tanvir et al., 2019) and can contribute to the formation of secondary inorganic aerosols (Karagulian et al., 2015). Tropospheric trace gases, linked to urban, industrial, and agricultural activities, show the highest concentrations over Punjab (Jabeen and Khokhar, 2019, Zeb et al., 2019, Khokhar et al., 2015).

1.2.1. Lack of proper air quality monitoring networks in Pakistan

A major barrier to effective air quality management in Pakistan is the absence of a reliable, widespread air quality monitoring network, particularly for $\text{PM}_{2.5}$ concentrations. High capital and operational costs restrict the deployment of conventional monitoring stations, leading to reliance on outdated or insufficient equipment. Low-cost sensor networks offer a promising alternative for expanding coverage and improving air pollution surveillance.

1.2.2. Data scarcity and sharing

Another substantial challenge is to have a consistent and comprehensive database of all air quality parameters, source apportionment and emission inventories. Limited and inconsistent availability of reliable data has complicated the mitigation efforts. The lack of centralized, open-access repositories impedes informed decision-making and timely interventions. Therefore, establishing a standardized data-sharing framework is crucial for effective air quality management.

1.2.3. Lack of institutional capacity

Past efforts to install air quality monitoring stations, such as the JICA-funded program (2006–2009), failed due to inadequate technical expertise and financial constraints, leading to discontinued operations by 2014. The absence of skilled personnel to maintain monitoring systems highlights the need for capacity-building initiatives within environmental agencies. Maintenance and operation AQM stations is imperative with rectification of previous mistakes.

1.2.4. Trans-boundary air pollution

Pakistan shares a transboundary air shed with neighboring countries, exacerbating pollution through cross-border emissions. Agricultural burning in Indian Punjab contributes significantly to regional smog episodes (Aryal et al., 2021). The transboundary aspect of the problem is further exacerbated by the disparities in the quality and coverage of regulatory monitoring programs between countries as shown in Table 1. Publicly available air quality data can be accessed through the respective governmental portals: Pakistan (PAK-EPA), India (CPCB) and (OGD INDIA), Nepal (DOE NEPAL), and Bangladesh (DOE BANGLADESH). However, similar datasets are either limited or unavailable for Sri Lanka, Afghanistan, and the Maldives. This data asymmetry, coupled with ineffective implementation of agreements like the Malé Declaration limits regional cooperation (Khokhar et al., 2023). Strengthening cross-border air quality policies and data-sharing mechanisms is essential for long-term improvements.

Table 1

Distribution of government-owned air quality monitoring stations in South Asia along with the mean annual fine particulate matter exposure ($\mu\text{g}/\text{m}^3$).

Country	Mean Annual $\text{PM}_{2.5}$ Exposure ($\mu\text{g}/\text{m}^3$) (The World Bank, 2021)	No. of Government-Owned Stations	Locations	References
Pakistan	43	6	3 Lahore (Town Hall, Punjab University, Mobile Station), Rawalpindi, Islamabad and IESE- NUST*	(EPD, 2018)
Nepal	46	9	Nepalgunj, Lumbini, Dang, GBS Pokhara, Sauraha, Simara, Dhulikhel, Bhaktapur, Shankapark, Bharatpur, Hetauda, Bhimdatta (Mahendranagar), Pulchowk, DHM Pokhara, Surkhet, Dhankuta, Bhaishipati, Janakpur, Biratnagar, Rara, PU Pokhara, Damak, Ratnapark, Dhangadhi, Jhumka, TU Kritipur	(Public Health Update, 2021)
India	48	1213	28 States & 7 Union Territories	(PIB, 2022)
Bangladesh	42	11	Dhaka (D-Salam, Farmgate and S-Bhaban) Chittagong, Agrabab, Chittagong, Narayanganj, Gazipur, Khulna, Rajshahi, Sylhet and Barisal	(DoE, 2018, Rahman et al., 2019, Pal and Masum, 2021)
Sri Lanka	20	2	Battaramulla and Kandy	(CEA, 2013)
Maldives	12	8	HDh. Kulhudhuffushi, Sh. Funadhoo, Lh. Naifaru, Dh. Kudahuvadhoo, L. Gan, GDh. Thinadhoo, Fuvahmulah, Addu Hithadhoo	(Rehan, 2022)
Afghanistan	46	1	Kabul	(UNEP, 2019, Afghan et al., 2022)

*Since 2022.

1.2.5. Institutional and technological challenges

Pakistan's air quality management is hindered by limited government-run monitoring facilities, inadequate coordination among stakeholders, and a lack of structured feedback mechanisms. The absence of continuous air quality monitoring networks, high capital costs of conventional monitoring systems, and reliance on sporadic

satellite observations create significant data gaps (Anjum et al., 2021). A hybrid approach, integrating low-cost sensor networks with conventional monitoring stations, is crucial for improving air quality assessment.

Moreover, poor coordination between policymakers, industries, and the public exacerbates air pollution challenges in Pakistan. Weak institutional collaboration and ineffective public-private partnerships hinder research, policy implementation, and regulatory enforcement. Strengthening academic-industry-government networks could improve data sharing and drive innovation (Gul and Ahmad, 2012).

Moreover, ineffective feedback mechanisms and low public awareness further contribute to policy failures. Raising awareness among stakeholders and improving access to reliable air quality data can enhance environmental policies and encourage proactive measures against pollution (Anjum et al., 2021).

2. Low-cost sensors – a potential solution

2.1. Hybrid monitoring

Low-cost air quality sensors have become ubiquitous throughout the world with many companies offering sensor networks e.g., PurpleAir, Clarity, and TSI BlueSky to name a few. Recent work has been done on characterizing the precision and accuracy of low-cost PM sensors, and developing frameworks to calibrate and maintain sensors in the field (Johnson et al., 2018, Zheng et al., 2018). These sensors have also proven to be effective in characterization of air quality in South Asian countries (Madhwal et al., 2024) while providing a much better understanding of the spatial and temporal variability of air pollutants when integrated with regulatory monitors. This underscores the ability of hybrid monitoring networks employing low-cost sensors to offer a potential solution for low- and middle-income countries with inadequate monitoring facilities. As these are compact, easily replaceable, deployable at various locations with better spatial coverage (Rogulski, 2017), less expensive than the traditional instruments (Giordano et al., 2021a), and are easily operable. Though, (Ródenas García et al., 2022) reported systematic studies that found low-cost sensors to be adequate for qualitative air quality analysis, a regular field calibration with a reference instrument is highly advised. Low-cost sensors have many applications that include, academic research, monitoring of citizens' exposure especially vulnerable groups, source apportionment, traffic hotspot assessment besides enhancing the quality and quantity of AQM through their deployment with the existing conventional monitoring infrastructure (Lezina and Misuyev, 2022).

3. Material and methods

3.1. Instrumentation

In this paper, we showcase findings from a particular low-cost sensor, the TSI BlueSky, and validate it against a reference monitor (BAM) to assess its accuracy, precision, and usability. These results suggest that low-cost monitors such as the TSI BlueSky PM sensor could be a valuable addition to regulatory monitors in Pakistan.

3.1.1. Low-Cost PM Sensors: BlueSky (Model: 8143, TSI INC. Shoreview, MN 55126, USA)

The study involved use of TSI BlueSky (8143) sensors (Fig. 1) that are real-time outdoor and indoor PM_{2.5} detectors that also measure PM_C (coarse PM), relative humidity and temperature. These sensors are made up of a Sensirion (SPS30) PM component, micro-controller, and SD card (BlueSky User Manual, 2023). These smaller and low-weight devices are easily deployable and adjustable at any point or location with a waterproof exterior. Each low-cost sensor is connected through Wi-Fi and the PM data (recent and archived) is visualized through a cloud-based at the official TSI-Link website. A built-in SD card in each sensor provides an



Fig. 1. Low-cost sensor Model: 8143, from TSI INC. USA installed at one of the sites in Pakistan.

alternative data storage option. It operates on the scattering mechanism of laser light in which the number of particles is converted to mass concentrations. The BlueSky sensor operates on the principle of light scattering to detect and measure particulate matter (PM) in the air. A light source, such as a laser or light-emitting diode (LED), illuminates airborne particles as they pass through an enclosed chamber. The dispersed light from these particles is then detected by a photo detector, which analyzes the scattering pattern to determine particle concentration and size distribution. To ensure consistent and accurate readings, the sensor incorporates an integrated fan, which maintains a steady airflow through the measurement chamber. This feature enhances measurement reliability, particularly under high pollution conditions. Additionally, the sensor's robust design and factory calibration contribute to its long-term stability and low maintenance requirements, making it a reliable choice for air quality monitoring applications (Sensirion, 2025).

PM_{2.5} data can be extracted from TSI Link website cloud, where it can be visualized for up to three months (maximum) in history. The time averaging interval can be set manually and was set to 1-minute for the analyses discussed in this paper. The TSI cloud stores data for a 3-month period for each sensor. However, by utilizing the machine learning algorithms, data from all the sensors can be downloaded simultaneously in a single code run using a coded Application Programming Interface (API).

3.1.2. Operation

The sensor features a dome shape with a silver mesh for air intake and exhaust adjacent to the micro-USB power supply port and configuration button. All the sensors needed to be registered on the TSI Link website (TSI Inc, 2025) and then harmonized by collocating the sensors all together in one place for a specific period, averages were compared to the median values, and the correction factors were calculated, and applied to each sensor' settings option available on TSI Link dashboard. The sensors were configured by browsing the IP Address (192.168.4.1) by entering the Wi-Fi username and password in the settings given underneath the sensor. A shift of the button towards the operating position activates the sensor. The sensor indicates its status through the blinking of a green light one blink per second signifies a successful internet connection and data transmission to the cloud, two blinks indicate an incorrect Wi-Fi password and three blinks suggest an inability to transmit data to the cloud, likely due to the absence of an internet connection. Additionally, successful connection status can be verified through the TSI Link Dashboard, which displays all registered sensors associated with a specific representative account.

Representative account is setup on TSI Link and devices are registered with their serial numbers, names, locations, and calibration factors. The platform enables adjustments to settings, data sharing, relocation of devices, and visualization of historical data.

There are three methods for retrieving data from BlueSky devices:

1. Using TSI-provided scripts to download Quality Control (QC) and Quality Assured (QA) data in CSV format.
2. Downloading data directly from the TSI dashboard, which provides up to three months of data but lacks QA & QC.
3. Manually extracting data from the SD card, also without QA & QC.

Accessing data via the first and second methods requires an annual subscription for API access and downloads.

3.1.3. Time resolution and cost

BlueSky sensors provide real-time PM_{2.5} data at a 15-seconds time resolution in comparison to BAM which monitors after every 1-hour interval. Though, they cost 800 USD, in addition to a paid data services subscription.

3.1.4. Pre-deployment calibration

The median value of the total number of co-located sensors was compared with the data retrieved from each sensor. To get the best line of fit that shows the relationship between the sensor's readings and the median value, a linear regression analysis was performed on each sensor. The multiplicative harmonization factor for each sensor was calculated using the following equation after it concluded that a linear regression with zero intercepts was suitable:

$$\text{harmonization factor} = \frac{1}{\text{slope of the linear regression with zero intercept}}$$

3.1.5. BlueSky Low-Cost sensor network in Pakistan

A network of TSI BlueSky Low-cost sensor (LCS) network is established in Pakistan for real-time PM_{2.5} monitoring. A total of 61 LCSs were deployed in the whole country across various locations, including Lahore, Islamabad, and Rawalpindi (as shown in Fig. 2). These cities face numerous air quality challenges. Notably, Lahore has been ranked first on the list of the world's most polluted cities in 2022, and has seen a steady rise in PM_{2.5} levels (IQAir, 2022). The twin cities of Islamabad and Rawalpindi – having 36 of these total 61 strategically placed LCSs – are also experiencing increasing air quality concerns due to unprecedented increase in transportation and cottage industry.

In this paper, we discuss the validation of the BlueSky LCS which was carried out at the DG-Cement factory site in Chakwal, Pakistan. NUST PK-57 sensor was primarily deployed at this site because a reference monitor, Beta Attenuation Monitor (BAM), was available there. Additionally, data collected from these LCSs in 2023 is presented and analyzed in context of the pollution hotspots in Pakistan (as shown in Fig. 5).

3.2. Data Handling mechanisms

Data retrieved from low-cost sensors was handled using Python tools, for instance, 'matplotlib' and 'seaborn' Python libraries were used for data visualization and further analysis was performed employing the use of Microsoft Excel. Fig. 3 demonstrates calendar plots made after data retrieval from one of the LCSs. The color grading is done according to the US Air Quality Index (US AQI) for different pollution levels so it gives us a daily picture of the air quality situation in specific areas. The daily moving averages and weekly averages were also plotted to envisage further trends.

The 7-day moving average graph offers a dynamic perspective of the data, capturing short-term fluctuations. This makes it highly useful for real-time monitoring of PM_{2.5}, where the ability to respond quickly to changes is critical. In contrast, the weekly static average doesn't get

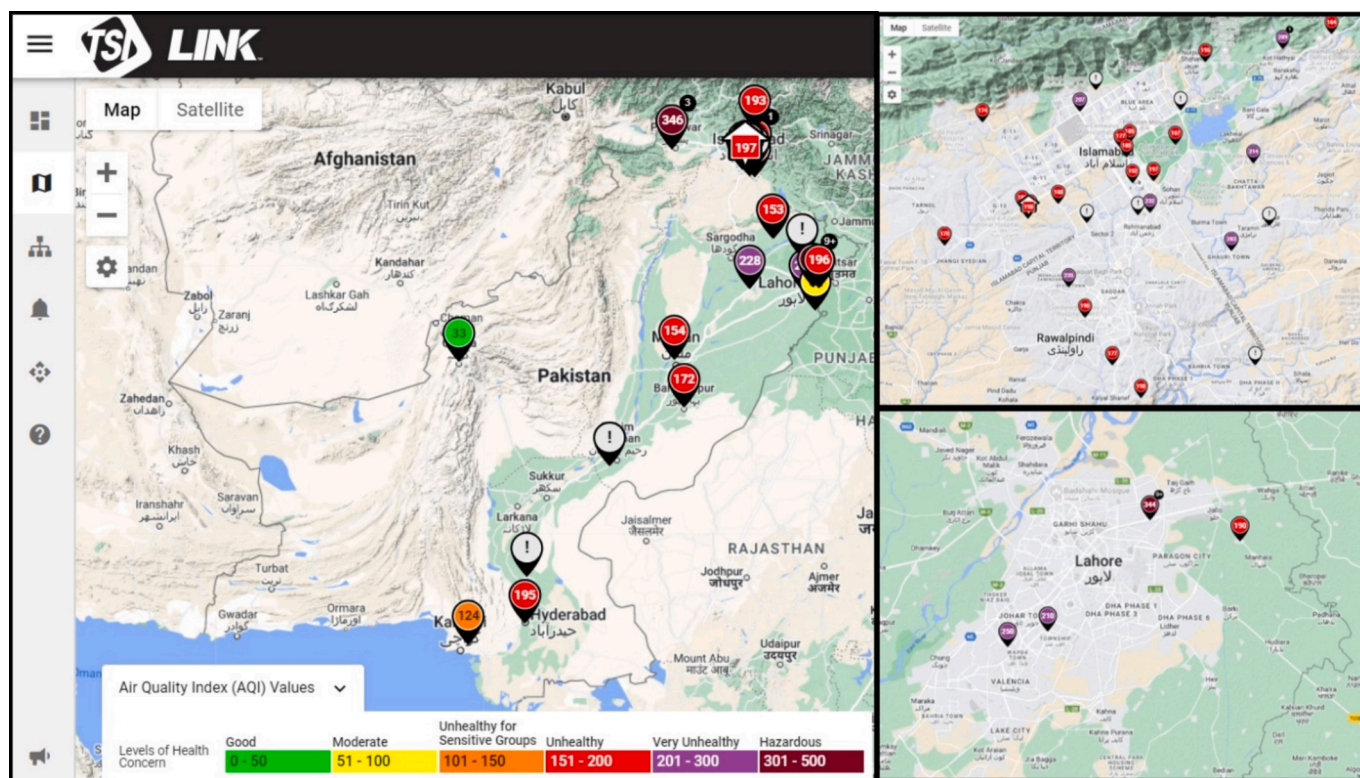


Fig. 2. Snapshot of BlueSky LCS Network established in Pakistan (Image).

Source: TSI Link Cloud

Calendar Chart

Data Set
Sample

From 01/06/2022
To 03/12/2023
Station Faisalabad PK34

Daily 7-Day Moving Average

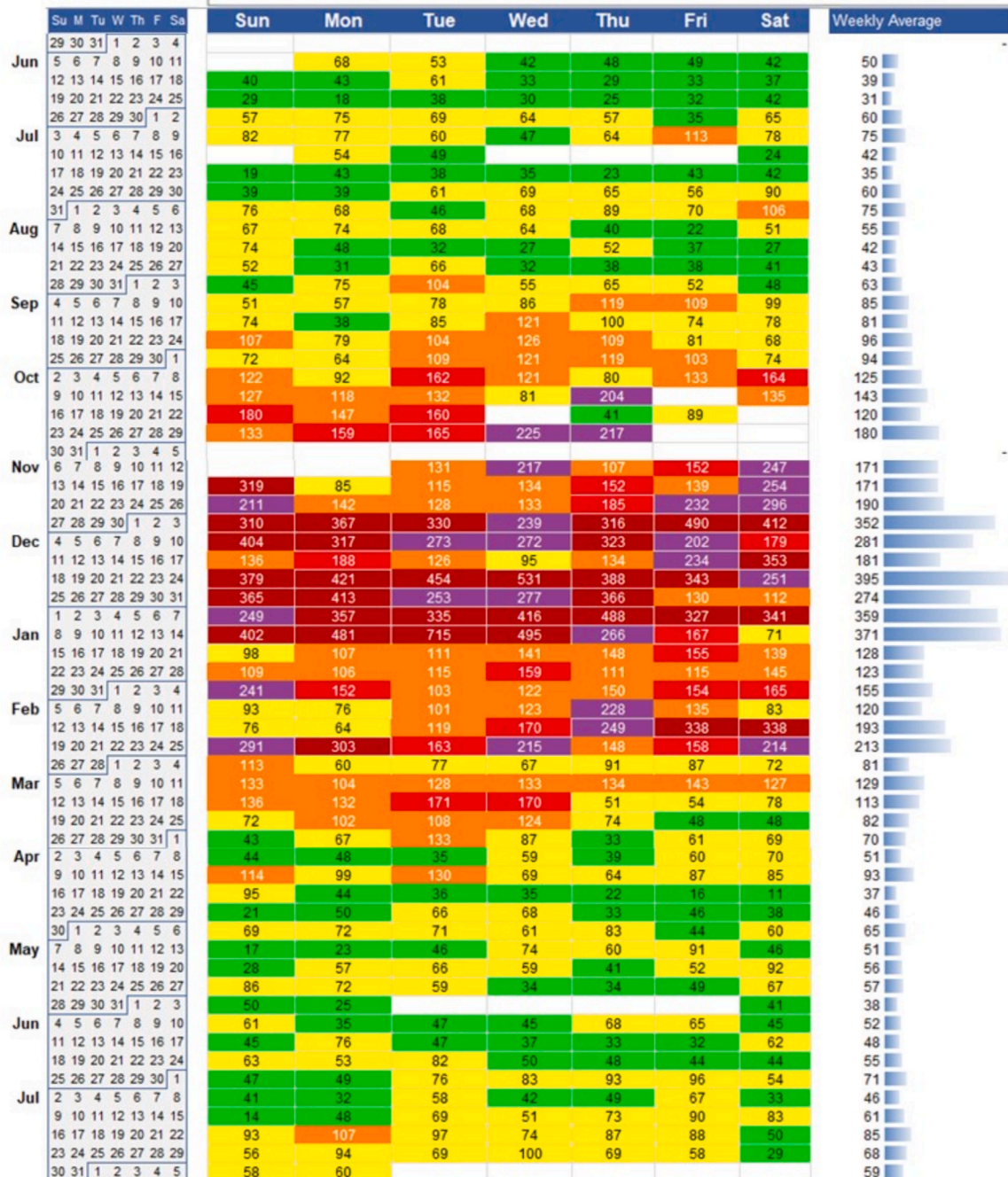
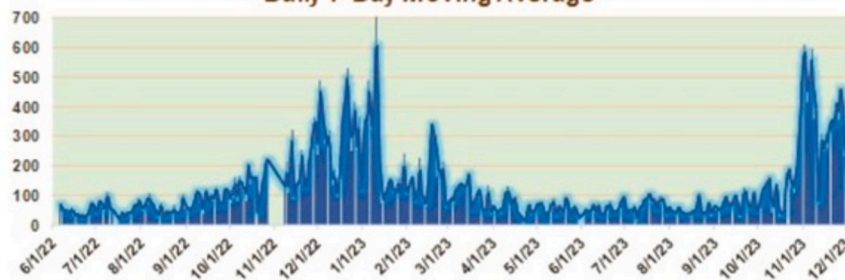


Fig. 3. Data Handling Mechanisms; calendar plots and 7-day moving average.

influenced by short-term changes and thus highlights a clear trend in $PM_{2.5}$ concentrations recorded over a set period of time. The calendar plot shown in Fig. 3 is a Gregorian calendar that is color-coded according to the US-EPA's Air Quality Index. This specific calendar plot is for PK34 –Faisalabad which gives the viewer an easy understanding of the air quality scenario in the city. Such kinds of analyses can help policy-makers and regulators to devise quick, effective and precise abatement measures even at micro levels.

4. Results and Discussions

4.1. Accuracy Estimations

4.1.1. BlueSky vs BAM Instrument

In order to validate the BlueSky sensors, comparisons were made with a reference Beta Attenuation Monitor (BAM) co-located at the DG cement factory in Chakwal, Pakistan.

To evaluate the performance and applicability of low-cost sensors for air quality monitoring, we analyzed the collected data using both adjusted and non-adjusted values. This comparative analysis enables a comprehensive understanding of the sensor's raw capabilities and the potential improvements that can be achieved through contextual

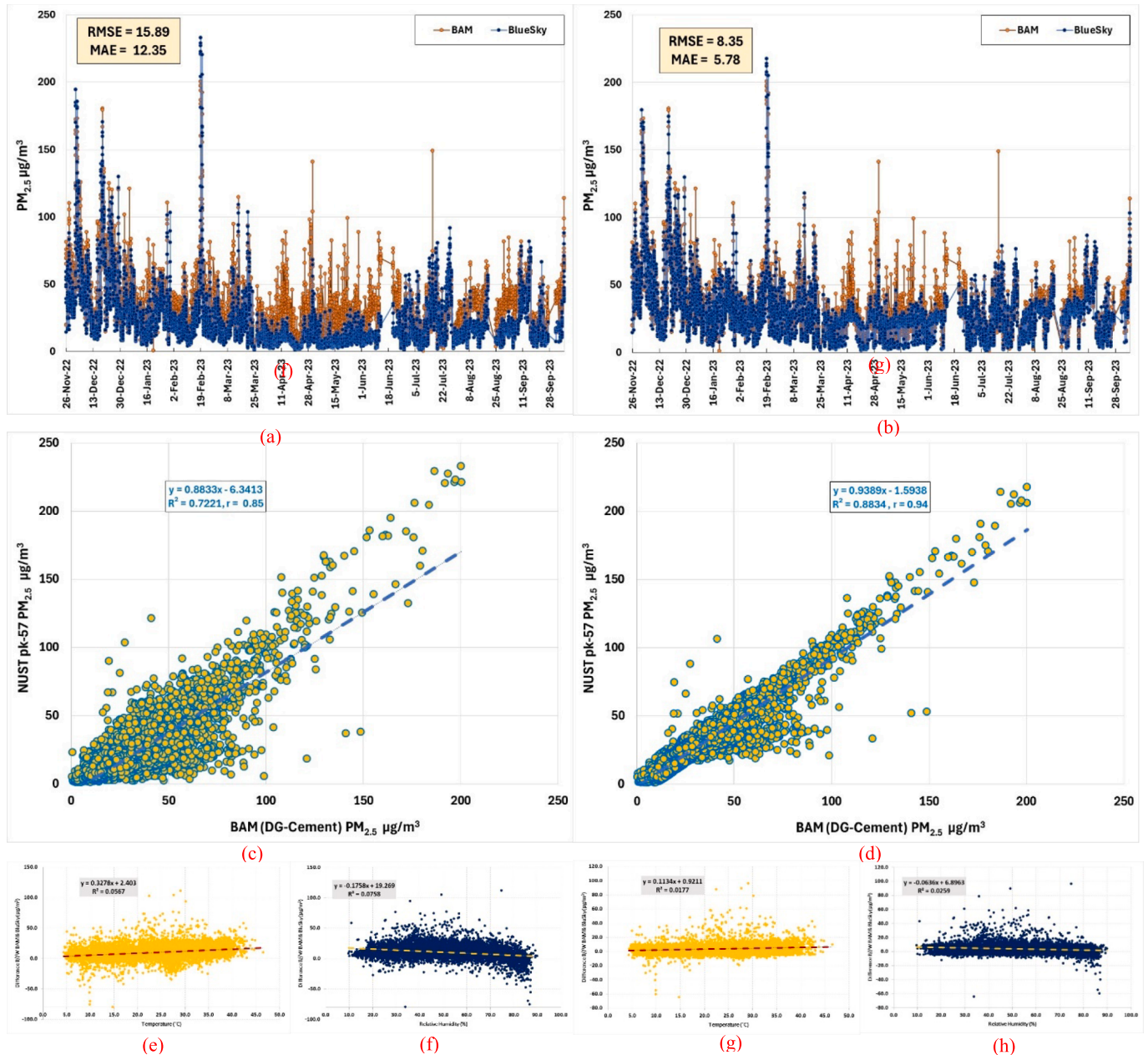


Fig. 4. (a) Non-adjusted hourly time-series plot of BlueSky (NUST PK-57) and BAM readings co-located at DG Cement, Chakwal, Pakistan (b) Adjusted hourly time-series plot comparing BlueSky (NUST PK-57) and BAM readings co-located at DG Cement, Chakwal, Pakistan (c) Scatter plot showing the hourly correlation between non-adjusted BlueSky (NUST PK-57) and BAM readings (d) Scatter plot showing the hourly correlation between adjusted BlueSky (NUST PK-57) and BAM readings (e) Correlation of temperature with the hourly difference between non-adjusted BlueSky and BAM readings (f) Correlation of relative humidity with the hourly difference between non-adjusted BlueSky and BAM readings (g) Correlation of temperature with the hourly difference between adjusted BlueSky and BAM readings (h) Correlation of relative humidity with the hourly difference between adjusted BlueSky and BAM readings.

calibration.

Firstly, the hourly time-series graph (Fig. 4a) plotted above compares the readings of the NUST PK-57 LCS with that of BAM readings. It shows that the BlueSky low-cost sensor effectively tracks the overall temporal trend of PM_{2.5} concentrations alongside the BAM. However, greater variability is observed, as highlighted in the corresponding scatterplot (Fig. 4c), where the RMSE and MAE rise to 16 µg/m³ and 12 µg/m³, respectively. The normalized RMSE (NRMSE) and normalized MAE (NMAE) increase to 8 % and 6 %. The R² value drops to 0.72, and the Pearson correlation to 0.85. Despite these increased error margins, the results clearly show that the BlueSky sensor can independently capture key air quality patterns, offering reliable performance for low-cost monitoring, particularly in resource-limited contexts where reference-grade instruments are not feasible.

In order to explore how sensor performance might be improved in a specific co-location scenario, a contextual statistical adjustment was applied to the BlueSky readings (Eq (1)). The purpose of employing its use is not to override or replace the original data, but to demonstrate the margin of enhancement possible when calibration is performed using concurrent measurements from a reference monitor.

$$Result_i = \begin{cases} B_i + Z \cdot \left(\frac{STDEV(B)}{STDEV(A)} \right) & \text{if } |B_i - A_i| > 10 \text{ and } B_i < A_i \\ B_i - Z \cdot \left(\frac{STDEV(B)}{STDEV(A)} \right) & \text{if } |B_i - A_i| > 10 \text{ and } B_i > A_i \\ B_i & \text{otherwise} \end{cases} \quad (1)$$

where Z is the inverse normal distribution function with a cumulative probability of 0.7413. STDEV(B) and STDEV(A) represent the standard deviations calculated for the entire columns B and A. B_i is BlueSky, NUST PK-57 while A_i is BAM colocated at DG Cement Chakwal site.

The adjusted data (Fig. 4b) showed a marked reduction in discrepancies, closely aligning with the BAM readings. As illustrated in the corresponding scatterplot (Fig. 4d), the RMSE and MAE dropped to 8 µg/m³ and 6 µg/m³, while the NRMSE and NMAE decreased to 4 % and 3 %. Subsequently, the R² value improved to 0.88, and the Pearson correlation rose to 0.94, indicating a strong linear relationship. These results demonstrate the potential for improving LCS accuracy through simple statistical adjustments when co-located with a reference monitor.

Nonetheless, it is important to emphasize that these adjustments are specific to this dataset and location, and may not be applicable across different environments or sensor deployments. The intention is not to suggest that non-adjusted data is inadequate but to highlight the significance of contextual calibration where higher precision is needed. In fact, despite the increase in error metrics and slightly reduced correlation with the use of unadjusted data, the BlueSky sensor still demonstrates substantial capability for monitoring PM_{2.5} levels. The unadjusted results provide a more genuine reflection of the LCS's performance, offering valuable insights into the real-world operation of these sensors without manipulation. This suggests that, even without adjustments, the BlueSky LCS remains a viable tool for air quality monitoring in resource-limited environments and the choice between using adjusted or non-adjusted data ultimately depends on the user's specific objectives and operational context.

For regulatory applications, daily averaged values of PM are utilized in the estimation of the Air Quality Index (AQI) to ensure compliance with air quality standards. Therefore, a daily-scale comparison between BAM and BlueSky data was conducted. This comparison led to a notable improvement in accuracy, as reflected by a significant reduction in RMSE from 8.35 to 5.42 and MAE 5.78 to 0.24 with adjusted values. Similarly, using non-adjusted values, the RMSE dropped from 16 to 1.14, and the MAE from 12 to 0.18. These findings suggest that daily averaging helps mitigate short-term fluctuations, enhancing the reliability of BlueSky measurements for air quality assessments and compliance.

The influence of meteorological variables on measurement discrepancies was also investigated. Fig. 4(e) and 4(g) illustrate the differences between BAM and BlueSky readings in relation to temperature, using non-adjusted and adjusted data, respectively. Both exhibit weak correlations, with R² values of 0.0567 (non-adjusted) and 0.0177 (adjusted), indicating minimal temperature influence. Similarly, Fig. 4(f) and 4(h) show the differences plotted against relative humidity, yielding R² values of 0.0758 (non-adjusted) and 0.0259 (adjusted), suggesting a slightly stronger—but still weak—association in the unadjusted dataset. Although literature suggests that humidity may affect PM_{2.5} readings due to hygroscopic particle growth, the observed correlations remain marginal in this dataset. These findings highlight that other meteorological parameters such as wind speed and direction may have a greater influence on discrepancies and should be explored in future studies.

LCS are engineered to function efficiently even in harsh environments and should be calibrated and validated following established standard procedures outlined in the literature (Zheng et al., 2018). In this study, all the remaining sensors were also calibrated by co-locating them with the BAM validated sensor NUST PK-57 for a few hours. The validated sensor was brought back to NUST, Islamabad and all the sensors were calibrated with it before their deployment at specific sites. However, the field performance analysis highlighted sensor malfunctions due to dust accumulation, requiring the development of a cleaning and recalibration protocol. Results for the PK-4 sensor that experienced dust accumulation, showed that the post-cleansing sensor calibration did maintain accuracy, with an NRMSE of 0.033. Additionally, exposure to high humidity and extreme temperatures impacted sensor functionality, necessitating regular maintenance. The findings confirm that with proper calibration and maintenance, the operational life span of the sensors can be increased (Giordano et al., 2021b) and resultantly LCSs can serve as an effective, low-cost alternative for large-scale air quality monitoring in developing regions.

4.2. Current PM_{2.5} pollution scenario in Pakistan

The air quality in Pakistan's key metropolis is consistently exacerbating (Anwar et al., 2021) and as discussed earlier in section 1.2.1, a significant lack of air quality monitoring stations across the country makes the continuous monitoring of these high pollution levels quite difficult (Javed et al., 2019). This problem is primarily attributed to the high cost associated with conventional air quality monitors. Therefore, low-cost sensors are one such viable alternative that can revolutionize air quality monitoring in Pakistan resulting in informed policy-making and robust public health initiatives.

In this section, the yearly data coverage from the LCS network established all across Pakistan gives a clear picture of the present air quality situation. Fig. 5 highlights the annual average number of days in 2023 exceeding daily Pak-NEQS (Pakistan's National Environment Quality Standards) i.e., 35 µg/m³ and daily WHO limit i.e., 15 µg/m³. Based on this data, it can be inferred that among these 16 cities with substantially variable air quality, the highest number of days exceeding Pak-NEQS was recorded in Faisalabad (307 days) whereas Ali Abad Hunza had only one such day. Moreover, similar to Faisalabad, major cities like Sadiqabad, Multan, Gujranwala, Phalia, Bahawalpur and Lahore recorded a significant number of days surpassing Pak-NEQS, indicating persistent air quality challenges. According to this city-specific data, cities like Faisalabad, Sadiqabad, Multan, and Gujranwala consistently exhibit higher pollution levels that surpass both Pak-NEQS and WHO standards on multiple days throughout the year. It should be noted that the number of sensors deployed across these cities varies and the ranking accounts for the average of each city. This data emphasizes the critical need for implementing effective air quality management strategies and environmental policies to address these challenges and improve public health. The yearly city-wise concentrations of fine particulate matter given in Fig. 5 make it quite clear that, unlike other cities, Faisalabad, Sadiqabad, Multan, Lahore and

Number of days exceeding Daily Pak-NEQS (2023)	Number of days exceeding Daily WHO Guidelines (2023)	Cities	Yearly Average ($\mu\text{g}/\text{m}^3$) (2023)	PM _{2.5} Pollution Ranking
307	330	Faisalabad	140	1
212	307	Sadiqabad	67	2
164	315	Multan	65	3
144	238	Lahore	64	4
145	295	Bahawalpur	64	5
149	253	Gujranwala	52	6
146	279	Phalia	44	7
101	226	Peshawar	44	8
121	245	Pawalpindi	39	9
80	195	Islamabad	32	10
88	293	Quetta	30	11
79	172	Hyderabad	28	12
74	147	Kasur	22	13
41	204	Mansehra	22	14
48	100	Gilgit	20	15
1	21	Ali Abad Hunza	8	16

Fig. 5. Annual average number of days exceeding daily Pak-NEQS and WHO guidelines for PM_{2.5} concentration along with city-wise ranking and yearly average concentration of PM_{2.5} in $\mu\text{g}/\text{m}^3$.

Gujranwala exhibit the highest values for PM pollution. The concentration across the country ranges from as low as $8.34 \mu\text{g}/\text{m}^3$ in Ali Abad to as high as $140 \mu\text{g}/\text{m}^3$ in Faisalabad. It is important to note that the three major pollution hotspots are the heavily industrialized cities of Pakistan (Faiz et al., 2015, Abbas et al., 2022) and therefore, industrial emissions along with vehicular pollution largely contribute to the worsening air quality in these cities.

In particular – Faisalabad, which has recorded the highest fine PM pollution – was ranked fourth on the list of most polluted cities in the world in 2019 (FDA, 2023, IQAir, 2019). The concentration of $140 \mu\text{g}/\text{m}^3$ recorded in this study, is also 4 times higher than Pakistan's NEQS and about 10 times higher than the WHO guidelines. The city's rising pollution levels are largely attributed to its heavy industrial activity (Sabir et al., 2024), as it serves as a major hub for textiles and manufacturing (IQAir, 2025). Factories, textile mills, brick kilns, and chemical plants rely on low-quality fuels such as coal and biomass, releasing significant amounts of fine particulate matter into the air (Tabinda et al., 2020). In addition to this, outdated diesel-powered trucks, buses, and motorcycles contribute to unregulated vehicular emissions, further deteriorating the air quality (Javed et al., 2015). Unlike developed nations, Faisalabad lacks strict fuel regulations and vehicle emission standards, resulting in excessively high PM_{2.5} concentrations from both industrial and transport sectors (Niaz and Zhou, 2013). The highest pollution levels are experienced during the winter months (December–January) due to temperature inversions, which trap pollutants close to the ground, causing PM_{2.5} concentrations to frequently exceed $200 \mu\text{g}/\text{m}^3$ (IQAir, 2025).

Despite certain concerns associated with the use of low-cost sensors with regards to their limitations in terms of accuracy, multitude of data, lifetime and acceptability, the aforementioned findings prove that with proper validation and regular maintenance, these sensors can be a viable alternative to conventional air quality monitoring. However, it is further discussed in this section how these concerns can be addressed effectively based on scientific evidences.

4.2.1. Accuracy

Improving the sensitivity, stability, and operational lifetime of sensors before replacing them is the primary technological problem associated with their usage in air pollution monitoring (White et al., 2012).

Various commercially available low-cost sensors have a sensitivity of 0.1 to $1 \mu\text{g}/\text{m}^3$ for PM_{2.5} and detection ranges for particle size from 0.3 μm to 10 μm sufficient enough to detect fine particle pollution (Kelly et al., 2017). There is a substantial improvement in accuracy as a result of large-scale application and an increased number of users in this field (Kumar et al., 2015). TSI's Blue Sky Sensor has shown a Pearson correlation of 0.94 and NRMSE obtained is 4 % with a NMAE of 3 % indicating that the calibrated sensors are highly accurate and compare well against the BAM data.

4.2.2. Multitude of data

Handling, processing, and analyzing the vast amount of data collected from numerous sensors deployed at a detailed spatial resolution poses a substantial obstacle. The associated cost and required technical skills are crucial factors common to LCS used in both indoor and outdoor settings.

Setting up a sensor network can be done at a relatively low cost, but the effort needed to maintain it and process the data will likely offset the cost of the hardware. Field calibration costs should also be considered. However, it has been concluded that, with the data handling tools outlined in section 4 and the benefits of efficient data management, promoting LCS monitoring of air quality is warranted (Sá et al., 2022).

4.2.3. Lifetime

Another limitation is the relatively shorter lifetime of the sensors—estimated to be approximately two years (Kumar et al., 2015). The main reason behind this is drift which eventually shortens the lifespan of LCS and affects their long-term stability and performance (Tancev, 2021, Kim et al., 2022). The majority of specialists concur that dust accumulation and the degrading of the sensor's electrical components are the causes of drift in low-cost sensors (Zimmerman, 2022, Bulot et al., 2020). These must be replaced annually or biannually, though there are ongoing efforts to enhance their lifetime, that could lead to some viable solutions in the near future. In our case, we have developed a dedicated protocol in order to enhance the lifetime and accuracy of LCSs, for further details please refer to Saeed et al 2023 (Revision Submitted). After regular cleaning (manual) of LCS deployed at various sites, the sensors are working efficiently over the period of 2 years. After each cleaning cycle the sensors were validated against the LCS deployed at

IESE-NUST site which are calibrated/tuned according to slope calculated from the validation of LCS (NUST PK-57) co-located with BAM at DG Cement Chakwal site on monthly basis. Moreover, we have indigenously developed a calibration chamber in order to take into account humidity factor under controlled laboratory conditions. It is installed to analyze the effect of temperature and humidity conditions. We do the offset correction accordingly.

In order to address the concern of the costs associated with replacing LCSs, here's an approximate comparison between the BAM and LCSs Table 2.

4.2.4. Acceptability

The lack of acceptability stems from significant resistance from regulatory authorities, driven by both a lack of will and capacity. Several factors contribute to this lack of acceptability, including insufficient capacity within regulatory bodies, the absence of world-class certifications, and misconceptions about the accuracy of LCSs. Additionally, economic challenges present a hurdle to the acceptance of LCSs. The high cost of data administration, processing, and visualization, which often exceeds the cost of the sensor system itself, needs to be reduced. Furthermore, if the decision-makers and researchers are not ready to adopt this technology, new challenges will arise. Enhancing their acceptance can be achieved through widespread adoption by the general public, including environmental activists and ordinary individuals. Environmental activists can also serve as a pressure tool on regulatory authorities. However, to bring about a fundamental change in air pollution monitoring, there must be concurrent advancements in awareness, education, and technology.

Large scale deployment of LCS and access via wireless links across the internet offer immense potential to monitor and control the air quality in a new generation of buildings (Kumar and Morawska, 2013), giving inhabitants authority over their environments. Though low-cost sensors have few limitations but their use has been widely encouraged due to the number of advantages they offer (Ródenas García et al., 2022).

5. Policy-making smart solutions

The establishment of LCS network can be very effective in developing countries as any extra measures required can be implemented in microenvironments due to finer scale coverage of both spatial and temporal scales. Furthermore, additional measures can be taken employing the use of artificial intelligence tools (data analysis and visualization etc.) to get the most benefits and efficient use of resources. These smart solutions can better inform the government compliance agencies to take steps beforehand to ensure improved air quality and to take the air pollution violators accountable. All these will result in smart policy-making and enhance the monitoring capacity of the countries to ensure cost-effective mitigatory and corrective measures.

6. Conclusions

Low-cost sensors represent an emerging technology with significant potential for air quality monitoring. This article provides strong evidence of the reliability and adequacy of these sensors. For countries in South Asia, particularly developing nations like Pakistan, where air quality monitoring facilities are inadequate, LCSs offer a promising solution for the management of air quality. In light of the complex challenges associated with declining air quality in these regions, the deployment of LCSs is increasingly necessary. The results discussed above demonstrate the reliability and robustness of the TSI BlueSky low-cost sensor data, validated against a co-located BAM reference monitor at DG Cement Chakwal. Moreover, the use of machine learning tools has led to improvements in accuracy by simplifying the data analysis process, thereby providing researchers and policymakers with clear insights into data trends over time. This facilitates smart decision-making for air

Table 2

A comparison of the estimated costs of traditional monitor; BAM with the low-cost sensors over a decade. According to this decadal comparison, LCS is approximately 85 % cheaper, even after considering the usual 2 years replacement. However, in our case, the network is sustained for more than 3 years due to consistent maintenance and calibration protocols on a regular basis.

Overall	Traditional Monitor BAM	Low-Cost Sensors
Equipment Cost	~ 30,000 USD	800 USD (with annual data subscription)
Installation Cost	~ 150–200 USD (transportation and labor needed)	~ 5–10 USD (shipping to the locations)
Operational Cost (per year)	~ 50–100 (high electricity consumption and not portable)	~ 5–10 USD (portable and requires only 5 W ~ 5 V and internet connection)
Maintenance Cost (per year)	~ 150–200 USD (required calibration by authorized expert technician to maintain)	~ 10–15 USD (anyone can do cleansing for regular maintenance in case of malfunctioning)
Replacement Cost (LCS replaced every 2 years)	N/A	~ 800 USD × 5 = 4000 USD
Total Cost Over 10 years	~ 32,150—33,200 USD	~ 4800 – 5100 USD

quality control, even at a micro level. The novelty of this work lies in addressing the challenges faced by different regulators for conducting reference air quality monitoring, which is well-suited for resource constrained countries. It can be implemented and replicated in any part of the world. Moreover, low-cost sensors due to their compact design, ease of replacement and deployment at various locations, high resolution, ease of operation and lower capital cost than the conventional/reference instruments, are highly recommended to be incorporated into hybrid monitoring networks for widescale monitoring.

CRedit authorship contribution statement

Muneeba Shabbir: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Talha Saeed:** Investigation. **Ahmad Saleem:** Validation. **Parkash Bhawe:** Project administration. **Mike Bergin:** Writing – review & editing, Resources, Project administration, Funding acquisition. **Muhammad Fahim Khokhar:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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